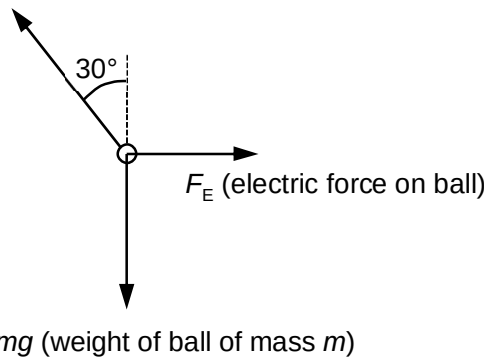
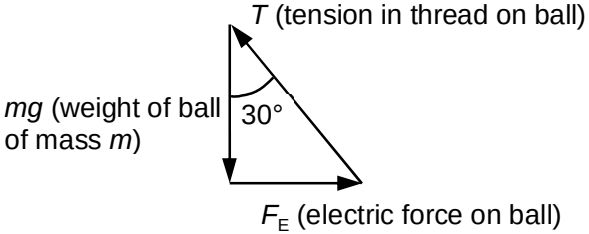


	longer time to reach it.
6(a)(i)	Electric field strength at a point is defined as the electric force exerted per unit positive charge placed at that point. <i>Note: The “unit positive charge” is necessary. Cannot omit either “unit” or “positive”.</i>
6(a)(ii)	Uniform electric field is a region of space in which a charge experiences a constant electric force.

No.	Solution
6(b)(i)	T (tension in thread on ball)  Vector triangle of the forces acting on ball: 
6(b)(ii)	Consider forces in equilibrium horizontally, $T \sin 30^\circ = qE$ --- (1) Consider forces in equilibrium vertically, $T \cos 30^\circ = mg$ --- (2) (1)/(2): $\tan 30^\circ = \frac{qE}{mg}$ $\frac{1}{\sqrt{3}} = \frac{qE}{mg}$ $E = \frac{mg}{\sqrt{3}q} \quad (\text{shown})$
6(b)(iii)	Using $E = \frac{mg}{\sqrt{3}q}$, $E = \frac{(10 \times 10^{-3})(9.81)}{\sqrt{3} \times 9.0 \times 10^{-9}}$ $= 6.293 \times 10^6$ $\approx 6.3 \times 10^6 \text{ N C}^{-1}$

No.	Solution
6(c)	The ball experiences a constant horizontal acceleration to the right due to the electric force, resulting in a horizontal velocity that is increasing uniformly to the right. At the same time, the ball experiences a vertical downward acceleration due to the gravitational force, resulting in a vertical velocity that is increasing uniformly downward. The resulting motion of the ball is along the straight line of action of the tension on the ball, at angle of 60° below the horizontal.
6(d)	The ball experiences a constant horizontal acceleration to the left due to the electric force, resulting in a horizontal velocity that is increasing uniformly to the left. The magnitude of the acceleration is smaller compared to part (c) because the magnitude of the charge on the ball is smaller. The motion of the ball is along a straight line at angle θ greater than 60° below the horizontal, as shown in the diagram below. 